$$\mathrm{f}(x) = \left(2 + \frac{kx}{8}\right)^7 \quad \text{where } k \text{ is a non-zero constant}$$

(a) Find the first 4 terms, in ascending powers of x, of the binomial expansion of $\mathrm{f}(x)$. Give each term in simplest form.

(4)

Given that, in the binomial expansion of $\mathrm{f}(x)$, the coefficients of x, x^2 and x^3 are the first 3 terms of an arithmetic progression,

(b) find, using algebra, the possible values of k.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(3)